# DESIGN REPORT: SMART SAKE TABLE

SMART SAKE TABLE

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# Executive Summary

Sake is a traditional Japanese alcoholic beverage; one step of sake's fermentation process involves the controlled growth of koji mold over rice. The current koji growth process at Void Sake Co. is time-consuming and labor intensive. The process relies on in-person qualitative analysis of the koji. If koji growth is left unchecked, it may overgrow or undergo a process called sporulation. Sporulation can be expedited by high temperatures or high humidity; it is also known to introduce unwanted flavors to the final sake product. Currently, each time koji is grown at The Void Sake Co. The head brewer must remain on the premises for the duration of the growth stage. This practice is unsustainable, given intense time and labor commitments.

The proposed design to monitor koji growth at The Void Sake Co. is a smart sake table. The smart sake table will reduce the amount of labor required during the koji growth stage. The design consists of six fans, paired with six temperature probes. The system aims to regulate the temperature of the koji bed without physical mixing. The design structure is based on the current koji table at Void Sake Co.

A dashboard mounted near the table allows the user to observe live readings for temperature, fan power, humidity, and table weight throughout the duration of the run. The user can use this live data to monitor and adjust the koji growth parameters, including physically mixing the rice bed if necessary. The design will reduce the amount of work required to grow the koji mold.

# Table of Contents

Executive Summary ..... 1
Introduction ..... 3
Purpose ..... 3
Research ..... 3
Relevant Standards ..... 3
Problem Statement ..... 4
Design Constraints ..... 4
Design Considerations and Criteria ..... 5
Proposed Solution ..... 5
Budget and Bill of Materials ..... 6
Conclusion and Recommendation(s) ..... 6
Conclusion ..... 6
Recommendation(s) ..... 6
References ..... 7
Appendix A: Sake Brewing and Koji Processing ..... 8
Appendix B: Koji Temperature Graph ..... 9

# Introduction

## Purpose

The process of fermentation is required in the production of alcoholic beverages. The sake, or Japanese rice wine, brewing is a unique process that is very labor intensive and artisanal, relying on both solid and liquid fermentation steps. In the solid fermentation step, a bed of rice is fermented using a traditional koji table. This table allows brewers to monitor and manipulate fungus growth. Currently, this process is labor-intensive, time-consuming, and difficult to standardize, as it depends on the subjective and qualitative analysis done by the brewer.

As the growth cycle of koji mold is in the brewer's control, deviations from allowed temperature and humidity ranges can lead to an undesired final product. A common problem with koji is the risk of sporulation, or overgrowth of the mold (Inoue, 2016). Sporulation is often signaled by a color change in the rice, which is why koji growth and its environmental conditions must constantly be monitored by the brewer. This color change signaling sporulation can be difficult to notice, often occurring within the bed of rice. The issue of sporulation can lead to waste and large expenses. As much of the current monitoring relies on visual cues, a solution that allows for consistent products is required to save both money and time.

## Research

Koji is a filamentous fungus, or a type of mold. However, the term koji also refers to rice that has already been inoculated with mold spores (Uwajimaya, 2025). The process for producing koji, which is used to create sake, originated in Japan and has become increasingly popular in the United States. In Japan, "sake" is a generic term to describe alcoholic beverages. When describing sake, the Japanese use the term "nihonshu" (Roussille, 2018). As seen in Figure A-1, the production of sake involves multiple steps. This includes rice steaming, koji making, fermentation, and pressing. For this project, the focus is on the koji making process. The most popular way to produce koji is by spore-inoculating a koji mold, *Aspergillus oryzae,* on a bed of steamed rice as outlined in Figure A-2. Koji produced this way is then suitable to produce fermented food products (Japan Patent No. 4115591, 1977).

Many important factors must be considered when designing a koji room. Insulation to prevent condensation, maintaining a set temperature and humidity, and ventilating to remove heat, water, and carbon dioxide from koji growth are critical design factors (Wong, 2021). The temperature-time curve gives insight into the different setpoints for the koji process as shown in Figure B-1. Due to spore growth and germination having different optimal temperatures, we cannot set a single temperature for the entire fermentation process (Wong, 2021). When looking at specific temperature growth curves, we see that there are no phases of high and low temperatures. Instead, there are specific points of growth where the koji bed requires high temperature, such as around five hours, and a need for low temperatures at approximately twenty-two hours (Inoue, 2016). Understanding these different constraints and desired ranges is important when working to improve the existing system.

## Relevant Standards

When working with koji to produce sake, there are a number of standards to adhere to ensure safety, reliability, and competency. Standard ANSI/ASAE D241.4 provides information surrounding the specific gravity, air space, and moisture content in rice kernels on a bulk basis. During the fermentation of koji, steamed rice absorbs water to allow for koji growth. Knowing these mass-moisture relationships for rice is important to ensure proper koji fermentation and to avoid sporulation. This standard will help the group to determine how the moisture content in the rice changes over time. Similarly, standard ASAE D243.5 provides data and equations to aid in estimating thermal properties for cereal grains, including rice. This standard has temperature ranges and thermal conductivity values for multiple styles

of rice. These values will help with creating setpoints for the temperature gauges that we plan to program during our testing and build phase, and for use in the operation of our project. Standard ANSI/ASAE S448.2 provides even more insight into the properties of rice. Since rice is the vessel in which koji is produced, the team found it important to gather multiple standards around its attributes. This standard is especially applicable as it is geared toward thin layers of grain. This standard lays out an experiment to measure the moisture content in rice and provides constants to use in these equations as well. As previously mentioned, the team is interested in the moisture content in the rice during fermentation in the koji room. This standard provides instruction on how the moisture content can be calculated.

The koji room has high humidity. When designing a new system for this space, one must consider the materials that can withstand these high levels of temperature and humidity. The EPA provides guidelines to follow to ensure that moisture is controlled. In EPA 402-F-13053, the EPA clarifies that moisture control does not require everything to be kept completely dry; vulnerable materials just need to be kept “dry enough”. Standard ASAE D243.5 discusses the wide range of digital thermometers that can be used in engineering applications. As the team continues to design our system, this information is valuable in deciding which thermometer to use when working in a koji room, keeping in mind the conditions of the room that the thermometer must be tolerant of. If the new additions to the table cannot withstand the environment of the koji room, the koji could become contaminated. Then, sake later produced would not be viable for sale and consumption. Ensuring that we have the proper equipment and materials to meet industry standards will be a crucial part of the planning process for this smart sake table project.

Another important aspect to consider is that food safety standards will have to be followed. There are specific requirements for materials and devices used in the production of consumable products, as highlighted by the FDA. Following their approved list of safe materials for encountering food during the inoculation and brewing process will be pertinent to ensuring the design of the system complies with federal standards (U.S. Food and Drug Administration, 2023).

## Problem Statement

Sake production is an artisanal Japanese practice growing in the USA. Traditional production relies on subjective in-person analysis, such as monitoring temperature, humidity, and color to prevent sporulation. Sporulation, the overgrowth of koji, leads to loss of product, more specifically 140kg of rice, 20 hours of labor, and additional lead times. A system to standardize control of koji growth without compromising the appearance of the current table would benefit the Void Sake Company.

## Design Constraints

This project requires consideration of various constraints to ensure that the design meets safety and reliability standards. One constraint in this design is space- the design solution must feet inside the 12’ x 10’ x 7’ foot room. The size of the table koji rice is made on must also be kept in mind, as it is 6’ by 4’ and has a table tray depth of 5”. Another facility constraint is the available power sources, as all plugs within and outside of the room operate on the same 110V, 60Hz, 20A breaker. There is also no Wi-Fi on site, so the design cannot utilize off-site remote control via an app. Looking at design constraints, the materials used in this project must work in a humid environment. More specifically, they will need to be moisture and rot resistant and food-grade. Finally, the traditional aspect of koji making must be respected, with the rice being mixed by hand.

## Design Considerations and Criteria

When designing the smart sake table, the cost, design, and process must be taken into consideration. The team has not been given a set budget but knows that the funding will be minimal. The design will also need to be easy to operate, as the facility does not have an engineer on-site. As previously mentioned, the traditional process must be respected, so the koji table will need minimal modification. This process requires manual labor and movement around the koji room, so reduced trip hazards must be considered to prevent the chances of bumps, trips, and falls.

The criteria selected for this smart sake table design is as follows:

- Record temperature of koji rice
- Record weight of rice
- Record humidity
- Automate fans
- Data visualization
- Easy operation
- Minimal human intervention

## Proposed Solution

The proposed design is a controlled drying table that helps koji rice dry evenly by managing airflow, humidity, and temperature inside the space. The table includes a raised perforated tray to hold the rice, six box fans that push air across the surface, and a dehumidifier to keep the moisture level within the proper range. Sensors such as thermocouples and load cells are built into the structure so the system can monitor temperature, dryness, and weight changes during the drying process. As shown in Figure 1, the exploded view highlights how the frame, multiple fan housings, load cells, and table components fit together to create a stable and easy-to-assemble system. This layout supports good airflow distribution, accessible wiring paths, and clear sensor placement, helping ensure the table performs consistently and can be controlled through the PLC. All of the sensor data will be sent to the PLC and then displayed on a connected monitor, where users can view temperature, humidity, and rice weight in real time. The system will also send the same information to an easy-to-use browser app, allowing users to check the status of the drying process from anywhere.

Figure 1: Exploded view of Koji Table

# Budget and Bill of Materials

## Bill of Materials

| Project: | Sake Table | | | | Academic Year: | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Team Members: | Benjamin Lin, Anastasia Myers, Makenna Hull, Natalie Cupples | | | | Revision of Form: | | | | 11/5/2025 | MH |
| System | Subsystem Nan | Assembly Sheet # | Item | Manufacturer | Manufacturer Part # | Qty | Unit | | Unit Cost | Total Price |
| System Name | Subsystem Name | Sheet number | CLICK PLUS PLC | AutomationDirect | C2-03CPU | 1 | Each | $ | 223.00 | $ 223.00 |
| System Name | Subsystem Name | Sheet number | CLICK PLUS PLC Relay | AutomationDirect | C0-08TR | 1 | Each | $ | 223.00 | $ 223.00 |
| System Name | Subsystem Name | Sheet number | CLICK PLUS PLC Thermo | AutomationDirect | C0-04THM | 2 | Each | $ | 217.00 | $ 434.00 |
| System Name | Subsystem Name | Sheet number | DIN Rail Power Supply | McMaster-Carr | 3718N117 | 1 | Each | $ | 68.14 | $ 68.14 |
| System Name | Subsystem Name | Sheet number | Type K Thermocouple | ProSense | THMK-N38P14-01 | 8 | Each | $ | 34.00 | $ 272.00 |
| System Name | Subsystem Name | Sheet number | Load Cells | AUTOSU | YZC-320C | 1 | Set of 4 | $ | 139.40 | $ 139.40 |
| System Name | Subsystem Name | Sheet number | Box Fan | Lasko | B20201 | 6 | Each | $ | 26.98 | $ 161.88 |
| System Name | Subsystem Name | Sheet number | 7" Touchscreen Display | Maple Systems Inc | HMI5070BV3 | 1 | Each | $ | 415.00 | $ 415.00 |
| System Name | Subsystem Name | Sheet number | Raspberry Pi 4 Model B | Raspberry Pi | SC0194(9) | 1 | Each | $ | 55.00 | $ 55.00 |
| System Name | Subsystem Name | Sheet number | 24" HDMI Monitor | LG Electronics | 24U41YA-B | 1 | Each | $ | 86.99 | $ 86.99 |
| System Name | Subsystem Name | Sheet number | Dehumidifier | Energy Star | VGE033A3BS | 1 | Each | $ | 269.99 | $ 269.99 |
| System Name | Subsystem Name | Sheet number | Thermal Imaging Modu | FLIR | 500-0763-01 | 1 | Each | $ | 179.95 | $ 179.95 |
| System Name | Subsystem Name | Sheet number | Humidity and Temperat | DFRobot | SEN0137 | 1 | Each | $ | 8.90 | $ 8.90 |
| System Name | Subsystem Name | Sheet number | Power Relay | Omron Automation and Sa | MY4N-GS-R DC24 | 2 | Each | $ | 12.36 | $ 24.72 |
| System Name | Subsystem Name | Sheet number | HDMI Cable | Raspberry Pi | SC0546 | 1 | Each | $ | 5.00 | $ 5.00 |
| | | | | | | | Total Cost | | | $ 2,566.97 |

# Conclusion and Recommendation(s)

## Conclusion

The current koji growth method at Void Sake Co. Is time-consuming, labor intensive. To prevent sporulation of the koji, this process relies heavily on in-person qualitative analysis of the koji growth and hand mixing to reduce rice bed temperature. However, the introduction of a smart sake table will reduce the amount of labor associated with each koji batch. The smart sake table will monitor and automatically remediate extreme rice temperatures, allowing the user to spend more time away from the run.

The system utilizes temperature probes to track temperature, paired with fans mounted underneath the bed to lower the bed temperature if needed. The included dashboard communicates live stats including temperature, humidity and power levels of the fans. The design is meant to conform to the current Void Sake Co. koji table, as well as omitting an automatic mixing component.

## Recommendation(s)

The current koji growth method at Void Sake Co. is heavily dependent on subjective, in-person analysis and is extremely labor-intensive. This reliance risks sporulation which results in lost product, labor hours, and extended lead times.

The proposed Smart Sake Table design is the necessary solution to standardize and automate the growth environment. The system, utilizing fans, a dehumidifier, and temperature probes, actively monitors and regulates the rice bed temperature without requiring physical mixing. Live data on temperature, humidity, and fan power is displayed on a nearby dashboard, replacing the need for constant, qualitative, human observation.

Based on the evidence that the Smart Sake Table effectively reduces the primary risks and labor constraints of the current process, immediate approval for the manufacturing and installation of the system is strongly recommended. This step will ensure product consistency, prevent costly batch losses, and significantly reduce the brewer's required time commitment.

# References

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# Appendix A: Sake Brewing and Koji Processing

The sake brewing process is a multi-stage process, often starting from polished rice to product distribution. In this process, the key point of interest is the koji making step. Koji starts with the addition of freshly steamed rice and ends with arresting the growth of koji on rice and moving the medium to a fermentation vessel. Figure A-1 shows the overall key steps in sake production.

Figure A-1. Overview Process Flow Chart for Sake Brewing

Figure A-2 shows a more detailed view of the koji inoculation process. The steamed rice intended for koji inoculation is spread evenly onto the table to cool. Once cold enough, spores are shaken onto the bed of rice and mixed in by hand. Following spore inoculation, the rice is then packed into a pile and covered to promote initial koji mycelium growth, allowing for the koji to establish itself in the grains for the rest of the process. Finally, it is spread out and mixed following specific temperatures and humidity to produce enzymes.

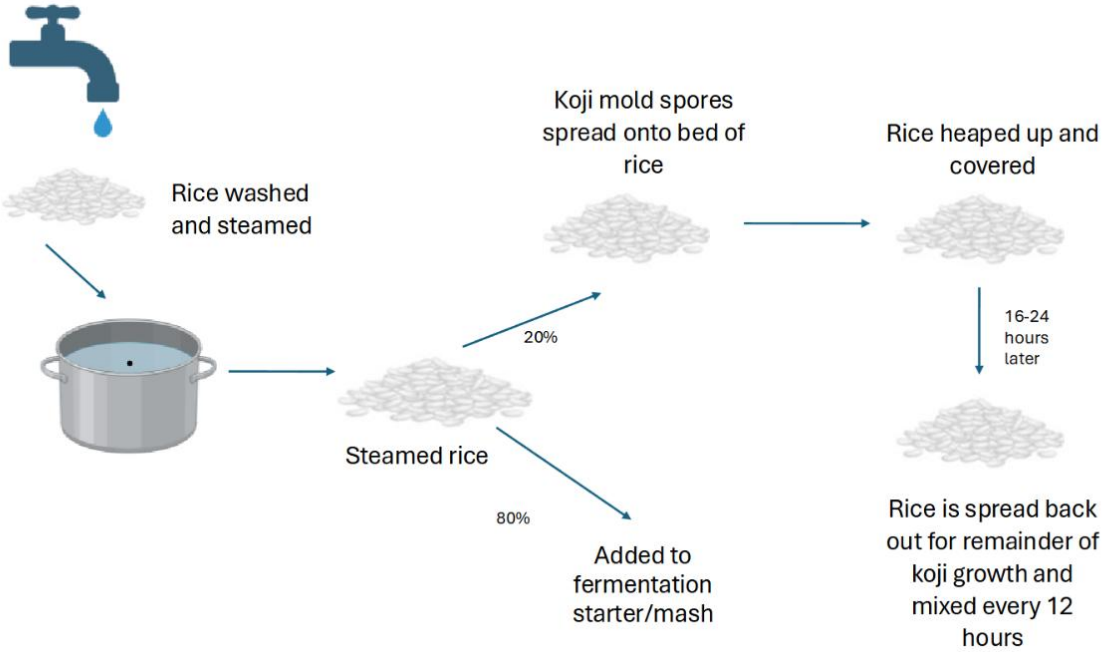


Figure A-2 Koji Inoculation

# Appendix B: Koji Temperature Graph

Koji inoculation follows intentional changes in temperature over time. The purpose of changing the environment is to support production of specific enzymes, such as amylases and proteases. Producing these enzymes is an essential component of koji, as it is used in fermentation for breaking down rice grains into simple sugars. The high temperature hold is also used to arrest the koji growth, keeping it from using energy to produce spores, which lowers the sugar concentration and introduces off tastes into the product.

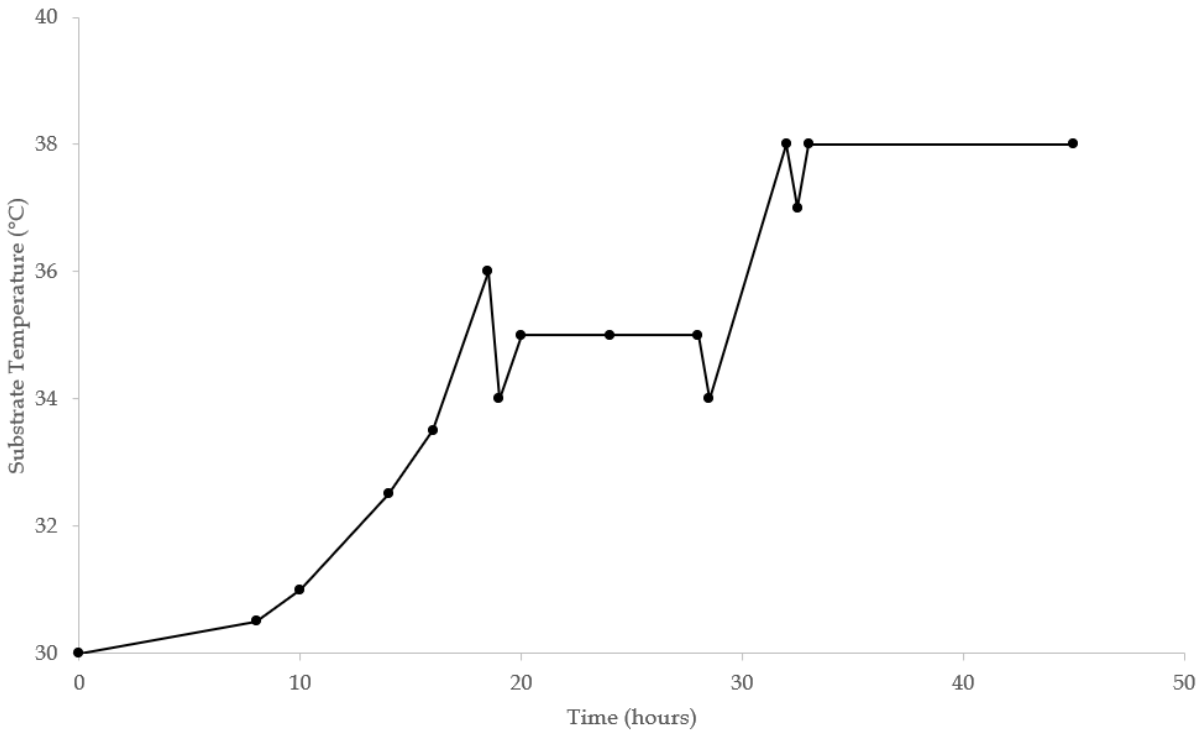


Figure B-1. Koji Temperature Graph for Inoculation